Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
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Digital Learning Handout

Part A: Content Design

Course Title	API-driven Cloud Native Solutions
Course No(s)	CC* ZG506 / SE* ZG506
Credit Units	5
Credit Model	3-1-1
Course Author	Prof. Shreyas Rao
Version No:	1.1
Date:	25/02/2025

Course Description:
Analyze, Design, Develop and Deploy cloud native applications in innovative areas such as Artificial Intelligence/Machine Learning (AI/ML), IoT, Data Analytics etc.; Build an end to end complex application; Extensive usage of well-known PaaS/APIs; Demonstration of compliance with relevant, industry adapted best practices; Deployment using modern strategies; Presentation of the Milestones and outcomes in appropriate forms; Periodic review of progress of the project by faculty.

Course Objectives

No	Course Objective
CO1	Explain the fundamentals of APIs, including RESTful and GraphQL APIs, and demonstrate how to design, specify, and implement API versioning
CO2	Identify the key components of the cloud-native ecosystem such as DevOps, microservices, containers, and serverless computing
CO3	Apply APIs and relevant tools to execute end-to-end data science and machine learning projects
CO4	Design AI assistants using APIs for use cases in natural language processing, speech recognition, and computer vision, and construct generative AI assistants using the RAG approach based on suitable language models
CO5	Develop and deploy end-to-end IoT and data analytics projects using APIs, serverless functions, and containers

Reference Book(s) & other resources:

R1	"Python for Everybody: Exploring Data Using Python 3", by Dr. Charles R. Severance, 2020 [Open Source under Creative Common License] URL - https://do1.dr-chuck.com/pythonlearn/EN_us/pythonlearn.pdf
R2	"Mastering API Architecture: Design, Operate, and Evolve API-Based Systems", by James Gough, Daniel Bryant and Matthew Auburn. Publisher: O'Reilly, 2023.
R3	"Cloud Native Patterns", by Cornelia Davis. Publisher: Manning, 2019

R4	"Introducing MLOps: How to scale Machine Learning in the Enterprise", by Mark Traveil. Publisher: O'Reilly, 2021
R5	Web Resources https://huggingface.co/ https://www.langchain.com/ https://docs.ragas.io/en/latest/concepts/metrics/index.html https://kubernetes.io/ https://www.docker.com/ https://hub.docker.com/ https://landscape.cncf.io/ https://flowiseai.com/ https://medium.com/credera-engineering/build-a-simple-rag-chatbot-with-langchain-b96b233e1b2a

Learning Outcomes: Students will be able to

LO1	Understand the fundamentals of APIs; design and build RESTful and GraphQL APIs
LO2	Understand the components of the cloud-native ecosystem, which includes DevOps, microservices, containers, and serverless computing
LO3	Execute end-to-end data science and machine learning projects, utilizing relevant APIs and tools
LO4	Design and develop AI applications using various APIs for natural language processing, speech recognition, and computer vision, as well as build generative AI assistants using the RAG approach and suitable language models
LO5	Implement end-to-end IoT and data analytics projects using APIs, serverless functions, and containers, leveraging relevant tools

Modular Content Structure

1. Python Programming [Self Study – Ref. Book R1]

- Structure of Python program
- Variables, Expressions and Statements
- Conditional Execution
- Functions
- Iteration
- Strings
- Files
- Lists
- Dictionaries
- Tuples
- Object-oriented programming
- Using databases and SQL
- Visualizing data

2. API Basics

- Introduction to APIs
- Design, Specify, and Build APIs

- RESTful APIs: Standards and Structure
- Specifying REST APIs using OpenAPI standard
- OpenAPI: Specification, Mocking, Semantic Versioning, and Tools
- OpenAPI Generators
- REST, GraphQL and gRPC based APIs
- API Versioning, Semantic versioning
- Examples of API - Google Maps API, Rapid API etc

3. Cloud Native Application

- Modern application requirements
- Cloud-native evolution
- Introducing Cloud-native software
- Cloud-enabled vs Cloud-based vs Cloud-native apps
- Examples for Cloud-native apps
- CNCF Landscape
- Overview of Cloud-native ecosystem
- DevOps and GitOps
  - Microservices and Service Mesh
  - Containers and Kubernetes
  - Serverless Computing and Serverless Stack
- Case Study: Analysis of a cloud-native application architecture

4. Data Science and Machine Learning

- Big Data and Characteristics
- Introduction to Data Science and Machine Learning
- Data Science Process
- Machine Learning Lifecycle
- Building an API driven Cloud-native Data Pipeline
  - Data ingestion
  - (Pre) Processing
  - Storage
  - Integration
  - Monitoring and Alerting
  - Scalability and Reliability
- API-driven ML pipelines
  - Model Development and Training
  - Model Deployment
  - Monitoring
  - Scalability
- MLOps practices
- Tools review
- Case Study: API-driven ML Model deployment

5. AI and Cognitive Services

Part-A: Basic APIs for AI applications

- Hugging Face APIs
- Computer Vision APIs
- Natural Language Processing APIs
- Speech Recognition APIs

- Case Study: API usage in AI Applications

Part-B: Language Models

- What are Language Models?
- LM Types
  - Large Language Model (LLM)
  - Small Language Model (SLM)
  - MMLU (Massive Multitask Language Understanding)
- LLMOps
- LangChain
- RAG (Retrieval Augmented Generation)
  - Document Loading
  - Document Splitting
  - Vectorstores and Embedding
  - Retrieval mechanisms
  - Question Answering
  - Chatbot – design and development
  - RAG Metrics – Deflection rate, Context precision, Context Relevance
- List of tools / APIs for LLMs
- Case Study: Real-world LLM application scenarios

Part-C: Cloud Native Application Deployment

- Dockers
- Containers
- Kubernetes
- Deployment Strategies

6. IoT and Data Analytics

- Internet of Things
- API integration for device communication
- Overview of Data Analytics
- Types of Data Analytics
- APIs for data ingestion, (pre)processing, analytics, and monitoring in IoT setup
- Serverless Functions in IoT environments
- Tools overview
- Case Study: API-driven data analytics in IoT projects

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	**API Basics**	• Introduction to APIs • Design, Specify, and Build APIs • RESTful APIs: Standards and Structure • Specifying REST APIs using OpenAPI standard • OpenAPI: Specification, Mocking, Semantic Versioning, and Tools • OpenAPI Generators • REST, GraphQL and gRPC based APIs	R2 - Chapter 1 Web Resources Lecture Notes

		• API Versioning, Semantic versioning • Examples of API - Google Maps API, Rapid API etc	
2	**Cloud Native Application**	• Modern application requirements • Cloud-native evolution • Introducing Cloud-native software • Cloud-enabled vs Cloud-based vs Cloud-native apps • Examples for Cloud-native apps	R2 - Chapter 1 R3 - Chapter 1 Web Resources Lecture Notes
3	**Cloud Native Application**	• CNCF Landscape • Overview of Cloud-native ecosystem ○ DevOps and GitOps ○ Microservices and Service Mesh ○ Containers and Kubernetes ○ Serverless Computing and Serverless Stack • Case Study: Analysis of a cloud-native application architecture	R3 - Chapter 1 Web Resources Lecture Notes
4	**Data Science and Machine Learning**	• Big Data and Characteristics • Introduction to Data Science and Machine Learning • Data Science Process • Machine Learning Lifecycle	Web Resources Lecture Notes
5	**Data Science and Machine Learning**	• Building an API driven Cloud-native Data Pipeline ○ Data ingestion ○ (Pre) Processing ○ Storage ○ Integration ○ Monitoring and Alerting ○ Scalability and Reliability • Demo	Web Resources Lecture Notes
6	**Data Science and Machine Learning**	• API-driven ML pipelines ○ Model Development and Training ○ Model Deployment ○ Monitoring ○ Scalability • MLOps practices	Web Resources Lecture Notes
7	**Data Science and Machine Learning**	• Tools review • Case Study: API-driven ML Model deployment • Demo	Web Resources Lecture Notes
8	**AI and Cognitive Services** **Part-A: Basic APIs for AI applications**	• Hugging Face APIs • Computer Vision APIs • Natural Language Processing APIs • Speech Recognition APIs	Web Resources Lecture Notes

		• Case Study: API usage in AI Applications	
9	**AI and Cognitive Services** **Part-B: Language Models**	• What are Language Models? • LM Types ○ Large Language Model (LLM) ○ Small Language Model (SLM) ○ MMLU (Massive Multitask Language Understanding) • LLMOps • LangChain	Web Resources Lecture Notes
10 & 11	**AI and Cognitive Services** **Part-B: Language Models**	• RAG (Retrieval Augmented Generation) ○ Document Loading ○ Document Splitting ○ Vectorstores and Embedding ○ Retrieval mechanisms ○ Question Answering ○ Chatbot – design and development ○ RAG Metrics – Deflection rate, Context precision, Context Relevance • List of tools / APIs for LLMs • Case Study: Real-world LLM application scenarios	Web Resources Lecture Notes
12	**AI and Cognitive Services** **Part-C: Cloud Native Application Deployment**	• Dockers • Containers • Kubernetes • Deployment Strategies	Web Resources Lecture Notes
13	**IoT and Data Analytics**	• Internet of Things • API integration for device communication • Overview of Data Analytics • Types of Data Analytics	Web Resources Lecture Notes
14 & 15	**IoT and Data Analytics**	• APIs ○ data ingestion ○ data (pre)processing ○ analytics ○ monitoring • Serverless Functions in IoT environments • Tools overview • Case Study: API-driven data analytics in IoT projects	Web Resources Lecture Notes
16	**Course Review and Discussion**	Review of Contact Sessions 1 to 16	

Experiential Learning Components:

1. Lab work: 5
2. Project work: 0
3. Case Study: 0
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of
a) API-based Data Pipeline with DataOps
b) API-based Machine Learning Pipeline with MLOps
c) Generative AI assistant to demonstrate NLP / CV/ Speech tasks

Scope of Experiential Learning Component:

Programming language - Python
Tools and libraries: Prefect, Prefect Cloud, MLFlow, AWS Sagemaker, HuggingFace APIs, Open AI APIs, LangChain, Amazon Bedrock

Lab Infrastructure:

Online/ Open source/ Virtual Lab/ Google Colab

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Demonstrate the design and development of an API-based data pipeline, which involves data ingestion, pre-processing, analysis and monitoring activities	5
2	Demonstrate the design and development of a Machine Learning pipeline which involves model development, training, deployment and monitoring activities	7
3	Demonstrate NLP tasks such as speech processing, language translation, text summarization, and text generation using Cloud-based APIs	8
4	Demonstrate a generative AI assistant, using the RAG approach, which answers queries about an application (chatbot)	11
5	Demonstrate an IoT and Data Analytics application	14

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1*	Quiz	Online	5%	1 week	To be announced
	Project / Assignment	Online	30%	30 days	To be announced
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	To be announced
EC - 3	Comprehensive Exam	Open Book	35%	2 ½ Hours	To be announced

EC1* (35%): Quiz: 5%, Project/Assignment: 30%
Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8
Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: https://elearn.bits-pilani.ac.in
Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: "open book" means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.
   EC3 Guidelines
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

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